

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 1 – Prompt Engineering Challenge**

Course Code:	CSA65	Course Title:	Generative AI and Large Scale Models
CO Assessed:	CO2 – Precision (BL1, BL2)	Assessment:	Assessment Tool 3 – Prompt Engineering Challenge
Weightage:	40% of CIA	Total Marks:	100 (2 Question)
CO1 Statement:	<i>Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems.(BL1, BL2)</i>		
Student Name:	N.Somashekar Naidu	Reg. No.:	192472347
Section / Batch:	CSE-AI	Date:	08-03-2026

Instructions to Students

- Answer 2 questions. Each question carries 50 Marks. Total: 100 Marks.
 - Write clear, well-structured prompts and explanations where asked.
 - Use your own original examples — do not copy from classmates or the internet verbatim.
 - Where a diagram or worked example is requested, label it clearly.
 - Illustrate prompting techniques (zero-shot, few-shot, chain-of-thought, role/system prompting, etc.) with concrete examples, not just definitions.
- Time allotted: 30 mins

Code of Conduct

I N.Somashekar Naidu(192472347)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.Somashekar Naidu_____

SECTION A –PROMPT ENGINEERING CHALLENGE**Q17. Comparative / Contrastive Evaluation Prompt**

Write a prompt that gives the model two pieces of writing (or two code snippets) and asks it to evaluate both against specific criteria, then declare a winner with justification for each criterion.

Aim

To use an API-based Large Language Model (LLM) to generate an initial response, critique its own answer based on predefined quality criteria, and then produce an improved final version through iterative self-reflection (Reflexion).

CODE:

```
from openai import OpenAI

client = OpenAI(
    api_key="sk-proj-Nmm1xmPfR2JbBq7_hpV94dLDJUM_DtDw1-
ioZea40r1nWeg9KtC6bF6L8b_oQmrGsHdM1354gFT3BlbkFJrcYwx4GunflHp3Me
j_UMbQRinVsBZPW5n9QE360CfD0kgiRpmo67LhwZiVH_nLw8Qu1ICdPPAA"
)

prompt = """
Task:
Write a 150-word article on the importance of Artificial Intelligence in Healthcare.

Step 1:
Generate the initial draft.

Step 2:
Critique your draft based on the following criteria:
1. Clarity
2. Grammar
3. Completeness
4. Readability
5. Organization

Step 3:
Revise the draft using your critique and provide the improved final version.
"""

response = client.responses.create(
    model="gpt-5.4-mini",
    input=prompt
)

print(response.output_text)
```

OUTPUT:

Step 1: Initial Draft

Artificial Intelligence is becoming an important part of healthcare. It helps doctors find diseases faster and more accurately by analyzing medical data, images, and patient records. AI tools can detect problems such as cancer, heart disease, and diabetes at an early stage, which improves treatment results. It also reduces the workload of medical staff by automating routine tasks like scheduling, record keeping, and monitoring patients. In addition, AI can support personalized medicine by suggesting treatments based on a patient's unique condition. This leads to better care and fewer mistakes. AI is also useful in hospitals for predicting patient needs and managing resources more efficiently. As healthcare continues to grow more complex, Artificial Intelligence will play a major role in improving efficiency, accuracy, and patient outcomes.

Step 2: Critique

- **1. Clarity:**** The draft is clear and easy to understand.
- **2. Grammar:**** Grammar is correct overall, with smooth sentence structure.
- **3. Completeness:**** It covers several key benefits, but could mention quicker decision-making or cost savings for more depth.
- **4. Readability:**** The wording is simple and readable, though slightly repetitive in places.
- **5. Organization:**** The ideas are logically arranged, moving from diagnosis to workload reduction to personalized care.

Step 3: Revised Final Version

Artificial Intelligence is transforming healthcare by making diagnosis, treatment, and hospital management more efficient. AI can quickly analyze medical images, test results, and patient records to detect diseases such as cancer, heart problems, and diabetes at earlier stages. This early detection helps doctors begin treatment sooner and improve patient outcomes. AI also reduces the burden on healthcare workers by handling routine tasks like appointment scheduling, documentation, and patient monitoring. In addition, it supports personalized medicine by recommending treatments

based on each patient's unique health data. Hospitals use AI to predict patient needs, manage resources, and reduce errors. By improving accuracy, saving time, and supporting better decisions, Artificial Intelligence is becoming essential in modern healthcare.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technique Selection	30%	Correctly applies zero-shot/one-shot/few-shot & CoT technique appropriate to each task	Applies techniques with minor mismatch to task	Limited understanding of which technique to use	Techniques applied incorrectly or not evident
Output Effectiveness	25%	Prompts produce accurate, task-appropriate outputs across all assigned tasks	Prompts work well for most tasks	Prompts work for only a few tasks	Prompts are largely ineffective
Prompt Craftsmanship	20%	Prompts are clear, specific, and reusable	Prompts are mostly clear	Prompts are vague or inconsistent	Prompts are poorly constructed
Prompt-Output Documentation	15%	Well-documented prompt-output pairs with clear rationale	Documented with minor gaps	Minimal documentation of prompts/outputs	No rationale provided for prompt choices

Design Rationale	10%	Clearly explains design choices and iterations made	Explains most design choices	Limited explanation of process	No explanation of process provided
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Marks Evaluation:

Question No.	Technique Selection (30%)	Output Effectiveness(25%)	Prompt Craftsmanship (20%)	Prompt-Output Documentation (15%)	Design Rationale(10%)	100
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____